

Topic 8 LMS Assessment

Specialist–Auditor Multi-Agent Quote Approval Workflow

n8n + Google Gemini

1. Project Overview

This project demonstrates a collaborative multi-agent workflow in n8n. Two specialized AI Agents perform separate responsibilities: the Evaluator analyzes and characterizes a quote, while the Auditor independently validates the quote against business rules and produces the final approval decision.

Workflow flow:

Mock Quote Request → Evaluator Agent → Structured Context Handoff → Auditor Agent → Auditor Decision / Fail-safe Validation → IF Routing → Approved Queue or Manual Review Queue

2. Assessment Objective

The implementation is designed to demonstrate separate evaluator and auditor responsibilities, structured communication between agents, business-rule validation, dynamic workflow routing, and successful end-to-end multi-agent collaboration.

3. Workflow Architecture

The exported n8n workflow contains the following functional components:

Node / Component	Purpose
Start	Manual workflow trigger.
Mock Quote Request	Creates structured quote input for testing.
Evaluator Agent	Pricing specialist that calculates totals, discount impact, and risk flags; it does not make the final approval decision.
Evaluator Gemini Chat Model	Google Gemini model used by the Evaluator.
Evaluator Output Schema	Enforces structured Evaluator output.
Context Handoff	Combines original quote data with Evaluator results for the Auditor.
Auditor Agent	Independently checks the quote against business rules and returns APPROVED or MANUAL_REVIEW.
Auditor Gemini Chat Model	Google Gemini model used by the Auditor.
Auditor Output Schema	Restricts Auditor decision to APPROVED or MANUAL_REVIEW and captures violations/reasons.
Auditor Decision	Deterministic fail-safe validation of the Auditor response and hard business rules.
Is Approved?	Routes based on final_decision.
Approved Queue	Receives approved quotes.
Manual Review Queue	Receives rejected/uncertain/invalid cases for human review.

4. Evaluator Agent

The Evaluator Agent is a pricing specialist. Its system instruction explicitly limits it to quote analysis and characterization. It computes the discounted total, reports effective discount, identifies risk flags, summarizes the evaluation, and provides a preliminary recommendation. It does not make the final approval decision.

Configuration

Model: models/gemini-3.1-flash-lite
Temperature: 0.0
Max Iterations: 5
Max Output Tokens: 512
Structured Output Parser: Enabled

5. Auditor Agent

The Auditor Agent is an independent compliance auditor. It receives the original quote plus the structured Evaluator output. It applies the configured business rules and returns exactly APPROVED or MANUAL_REVIEW. When uncertain or when data is missing or inconsistent, the prompt directs it to choose MANUAL_REVIEW.

Configuration

Model: models/gemini-3.1-flash-lite
Temperature: 0.0
Max Iterations: 5
Max Output Tokens: 512
Structured Output Parser: Enabled

6. Structured Context Handoff

The Context Handoff node passes the original quote fields together with Evaluator-generated fields, including evaluator recommendation, calculated total, effective discount, summary, and risk flags. This makes the collaboration between the two agents explicit and inspectable in n8n execution data.

7. Business Rules

The current workflow uses the following documented assumptions because the supplied Topic 8 Assessment Document does not specify numeric thresholds:

Rule	Configured condition
Discount	$\leq 20\%$
Quote value	$\leq 50,000$
Payment terms	Prepaid, Net 15, or Net 30
Quantity	> 0
Calculated total	Must match stated quote value within 1%

Important: these thresholds should be replaced only if the official Requirement Document or Implementation Guide specifies different values.

8. Fail-Safe Decision Validation

The Auditor Decision Code node normalizes the Auditor response, validates that the decision is one of the accepted values, and independently checks the hard business rules. Invalid or malformed Auditor output is routed to MANUAL_REVIEW. If the Auditor returns APPROVED while a hard rule fails, the deterministic guard overrides the decision to MANUAL_REVIEW. This prevents an unsafe AI approval from reaching the Approved Queue.

9. Dynamic Routing

The IF node checks:

```
{{ $json.final_decision }} equals APPROVED
```

TRUE → Approved Queue

FALSE → Manual Review Queue

10. Live Validation Results

Execution	Scenario	Observed result
155	Valid quote / rules pass	Real Gemini Evaluator → real Gemini Auditor → APPROVED → APPROVED_QUEUE
156	35% discount / rule violation	Real Gemini Evaluator → real Gemini Auditor → MANUAL_REVIEW → MANUAL_REVIEW_QUEUE

Both live executions completed successfully with real Google Gemini API calls. Structured parsers succeeded and no execution errors were reported.

11. Assessment Checklist

Requirement	Status
Mock Quote Request Configured	Verified
Evaluator Agent Added	Verified
Auditor Agent Added	Verified
Chat Models Connected	Verified
Temperature Set to 0.0	Verified
Max Tokens Configured	Verified
Max Iterations Set to 5	Verified
Evaluator Prompt Configured	Verified
Auditor Prompt Configured	Verified
Context Handoff Configured	Verified
Business Rules Implemented	Verified; thresholds documented as assumptions
Auditor Decision Verified	Verified
IF Routing Configured	Verified
Approved Queue Working	Verified in execution 155
Manual Review Queue Working	Verified in execution 156
Specialist-Auditor Workflow Completed	Verified

12. Submission Deliverables

The project is prepared for the required LMS submission package: Loom/YouTube demonstration, public GitHub repository, exported n8n workflow JSON, workflow documentation, screenshots, README, and implementation notes.

Recommended GitHub repository name: topic8-specialist-auditor-multi-agent-n8n

Recommended structure:

```
README.md
docs/workflow-documentation.md
docs/implementation-notes.md
docs/loom-script.md
```

```
docs/screenshots-checklist.md
docs/assessment-checklist.md
workflow/topic8-workflow.json
workflow/main.workflow.ts
screenshots/...
```

Demo video

Loom: <https://www.loom.com/share/3bb805d68290477e84a972500d76e5b5>

13. Security Note

The Google Gemini API credential is referenced by n8n, but the secret API key must not be included in GitHub files, screenshots, documentation, or the Loom recording.

14. Conclusion

The workflow demonstrates the required Specialist–Auditor multi-agent pattern with two independent Gemini-powered agents, structured context handoff, business-rule auditing, deterministic fail-safe validation, and dynamic routing to Approved and Manual Review queues. Live executions 155 and 156 verified both decision paths.